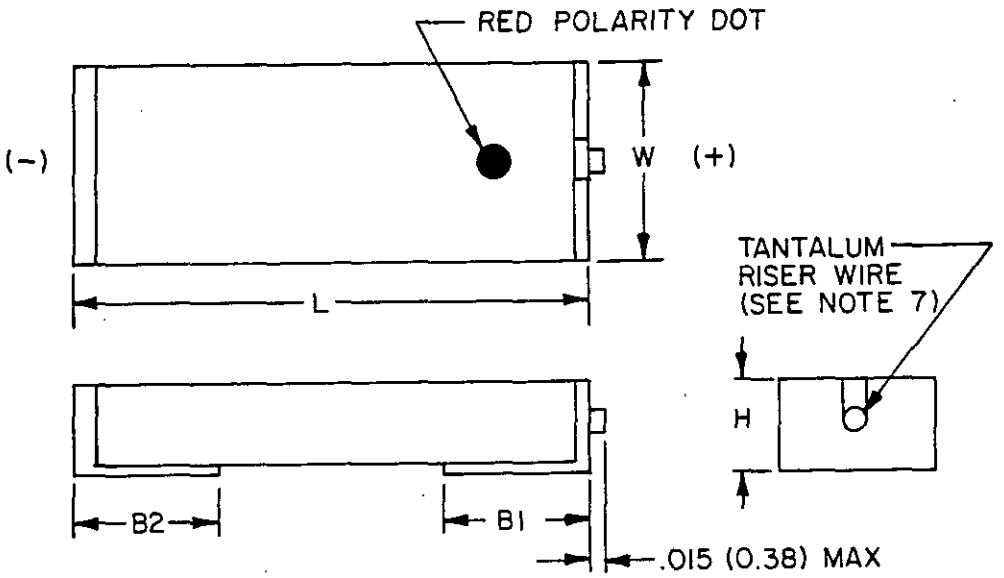
MILITARY SPECIFICATION SHEET

CAPACITORS, CHIP, FIXED, TANTALUM, ESTABLISHED RELIABILITY, STYLE CWR12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55365.



Dimensions					
Case code	(W) Width ±0.015 (0.38)	(L) Length ±0.015 (0.38)	(H) Height ±0.015 (0.38)	(B1) +Pad width ±0.010 (0.25)	(B2) -Pad width -.010, +.025 (-0.25, +0.64)
A	.100 (2.54)	.150 (3.81)	.050 (1.27)	.030 (0.76)	.030 (0.76)
B	.100 (2.54)	.200 (5.08)	.050 (1.27)	.030 (0.76)	.030 (0.76)
C	.135 (3.43)	.220 (5.59)	.070 (1.78)	.030 (0.76)	.030 (0.76)
D	.110 (2.79)	.265 (6.73)	.110 (2.79)	.050 (1.27)	.050 (1.27)
E	.160 (4.06)	.275 (6.98)	.110 (2.79)	.050 (1.27)	.050 (1.27)

FIGURE 1. Style CWR12 (conformally coated) capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses and are given for general information only.
3. For polarity identification: When chip is in mounting position, the weld stub is positive, and the top negative surface is conformally coated.
4. Anode riser (weld stub) may extend case size length dimension by .015 inch (0.38 mm) maximum.
5. These capacitors are designed for mounting by dip soldering, vapor phase reflow soldering, or by other conventional means.
6. Anode and cathode terminals may be two-sided, three-sided, or five-sided, and may not be the same on some designs, but shall meet all specification requirements.
7. Terminal-tantalum riser wire (+) weld area and terminal edges are not solderable.
8. The anode riser wire connection shall have a continuous weld of a minimum of 25 percent of the contact surface.

FIGURE 1. Style CWR12 (conformally coated) capacitors - Continued.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Chip.

Material: Sintered tantalum.

Leads: Unleaded.

Termination finish: In accordance with termination finish B or C of MIL-C-55365.

DC rated voltage: See table I. Above 85°C, voltage derating is required in accordance with MIL-C-55365.

Operating temperature range: -55°C to +125°C.

Failure rate level: In accordance with MIL-C-55365.

DC leakage (DCL): See table I.

Capacitance: See table I.

Capacitance tolerance: (J) ±5 percent, (K) ±10 percent, or (M) ±20 percent.

Dissipation factor (DF): See table I.

Equivalent series resistance (ESR) at 100 kHz: In accordance with MIL-C-55365. See table I herein.

Resistance to soldering heat: In accordance with MIL-C-55365.

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TABLE I. Style CWR12 characteristics.

PIN 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor			ESR max 100 kHz at 25°C	Case size
			at 25°C	at 85°C	at 125°C	at 25°C	at 85°C	at -55°C		
	Volts	µF	µA	µA	µA	%	%	%	Ohms	
CWR12C 107	4	100.0	4	40	48	10	12	12	0.5	C
CWR12C 187	4	180.0	8	80	96	10	12	12	0.5	D
CWR12C 227	4	220.0	9	90	108	10	12	12	0.5	E
CWR12D 336	6	33.0	2	20	24	10	12	12	1.0	A
CWR12D 476	6	47.0	3	30	36	10	12	12	0.8	B
CWR12D 686	6	68.0	4	40	48	10	12	12	0.7	C
CWR12D 107	6	100.0	6	60	72	10	12	12	0.5	D
CWR12D 127	6	120.0	8	80	96	10	12	12	0.5	D
CWR12D 157	6	150.0	9	90	108	10	12	12	0.5	D
CWR12D 187	6	180.0	11	110	132	10	12	12	0.5	E
CWR12F 226	10	22.0	3	30	36	10	12	12	1.0	A
CWR12F 476	10	47.0	5	50	60	10	12	12	0.7	C
CWR12F 686	10	68.0	7	70	84	10	12	12	0.7	D
CWR12F 107	10	100.0	10	100	120	10	12	12	0.5	E
CWR12F 127	10	120.0	12	120	144	10	12	12	0.5	E
CWR12H 106	15	10.0	2	20	24	6	8	8	1.5	A
CWR12H 156	15	15.0	3	30	36	6	8	8	1.5	A
CWR12H 336	15	33.0	5	50	60	6	8	8	0.8	C
CWR12H 476	15	47.0	8	80	96	6	8	8	0.8	D
CWR12H 686	15	68.0	11	110	132	6	8	8	0.8	E
CWR12J 685	20	6.8	2	20	24	6	8	8	2.5	A
CWR12J 106	20	10.0	2	20	24	6	8	8	2.5	B
CWR12J 226	20	22.0	5	50	60	6	8	8	1.0	C
CWR12J 336	20	33.0	7	70	84	6	8	8	1.0	D
CWR12J 476	20	47.0	10	100	120	6	8	8	1.0	E
CWR12K 335	25	3.3	1	10	12	6	8	8	3.5	A
CWR12K 475	25	4.7	2	20	24	6	8	8	2.5	A
CWR12K 685	25	6.8	2	20	24	6	8	8	2.0	B
CWR12K 106	25	10.0	3	30	36	6	8	8	2.0	C
CWR12K 156	25	15.0	4	40	48	6	8	8	1.0	C
CWR12K 226	25	22.0	6	60	72	6	8	8	1.0	D
CWR12K 336	25	33.0	9	90	108	6	8	8	1.0	E
CWR12M 105	35	1.0	1	10	12	6	8	8	6.0	A
CWR12M 155	35	1.5	1	10	12	6	8	8	5.0	A
CWR12M 225	35	2.2	1	10	12	6	8	8	5.0	A
CWR12M 335	35	3.3	2	20	24	6	8	8	2.5	B
CWR12M 475	35	4.7	2	20	24	6	8	8	2.5	C
CWR12M 685	35	6.8	3	30	36	6	8	8	2.0	C
CWR12M 106	35	10.0	4	40	48	6	8	8	2.0	D
CWR12M 156	35	15.0	6	60	72	6	8	8	2.0	E

See footnote at end of table.

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TABLE I. Style CWR12 characteristics - Continued.

PIN 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor			ESR max 100 kHz at 25°C	Case size
			at 25°C	at 85°C	at 125°C	at 25°C	at 85°C	at -55°C		
	Volts	µF	µA	µA	µA	%	%	%	Ohms	
CWR12N 104	50	0.1	1	10	12	6	8	8	20.0	A
CWR12N 154	50	0.15	1	10	12	6	8	8	20.0	A
CWR12N 224	50	0.22	1	10	12	6	8	8	15.0	A
CWR12N 334	50	0.33	1	10	12	6	8	8	15.0	A
CWR12N 474	50	0.47	1	10	12	6	8	8	10.0	A
CWR12N 684	50	0.68	1	10	12	6	8	8	8.0	A
CWR12N 105	50	1.0	1	10	12	6	8	8	6.0	A
CWR12N 155	50	1.5	1	10	12	6	8	8	5.0	A
CWR12N 225	50	2.2	2	20	24	6	8	8	3.5	B
CWR12N 335	50	3.3	2	20	24	6	8	8	2.5	C
CWR12N 475	50	4.7	3	30	36	6	8	8	2.5	D
CWR12N 685	50	6.8	4	40	48	6	8	8	2.5	D
CWR12N 106	50	10.0	5	50	60	6	8	8	2.5	E

1/ Complete PIN shall include additional symbols to indicate termination finish, capacitance tolerance, and failure rate level.

Stability at low and high temperatures: In accordance with MIL-C-55365.

Step 1 (+25°C):
DCL: See table I.
Cap.: Shall not exceed tolerance of table I value.
DF: See table I.
ESR: See table I.

Step 2 (-55°C):
ΔCap.: Shall not exceed ±10 percent of initial measurement.
DF: See table I.

Step 3 (+25°C):
DCL: See table I.
ΔCap.: Shall not exceed ±5 percent of initial +25°C measurement.
DF: See table I.

Step 4 (+85°C):
DCL: See table I.
ΔCap.: Shall not exceed ±10 percent of initial +25°C measurement.
DF: See table I.

Step 5 (+125°C):
DCL: See table I.
ΔCap.: Shall not exceed ±15 percent of initial +25°C measurement.
DF: See table I.

Step 6 (+25°C):
DCL: See table I.
ΔCap.: Shall not exceed ±5 percent of initial +25°C measurement.
DF: See table I.
ESR: See table I.

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Surge voltage: In accordance with MIL-C-55365.
DCL: See table I.
ΔCap.: Within ±5 percent of initial value.
DF: See table I.
ESR: See table I.

Life: Method 108 of MIL-STD-202. Exponential only.

2,000 hours:

At +25°C:
DCL: See table I.
ΔCap.: Shall not exceed ±5 percent of initial measured value.
DF: See table I.
ESR: See table I.

At +85°C:
DCL: See table I.

At +125°C:
DCL: See table I.

10,000 hours; Exponential only.

At +25°C:
DCL: 200 percent of initial requirement.
ΔCap.: Shall not exceed ±10 percent of initial measured value.
DF: See table I.
ESR: See table I.

At +85°C:
DCL: 200 percent of initial requirement.

At +125°C:
DCL: 200 percent of initial requirement.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - SH
Air Force - 17, 99
NASA - NA
DLA - ES

User activities:
Army - AR
Navy - AS, CG, MC, OS
Air Force - 11, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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